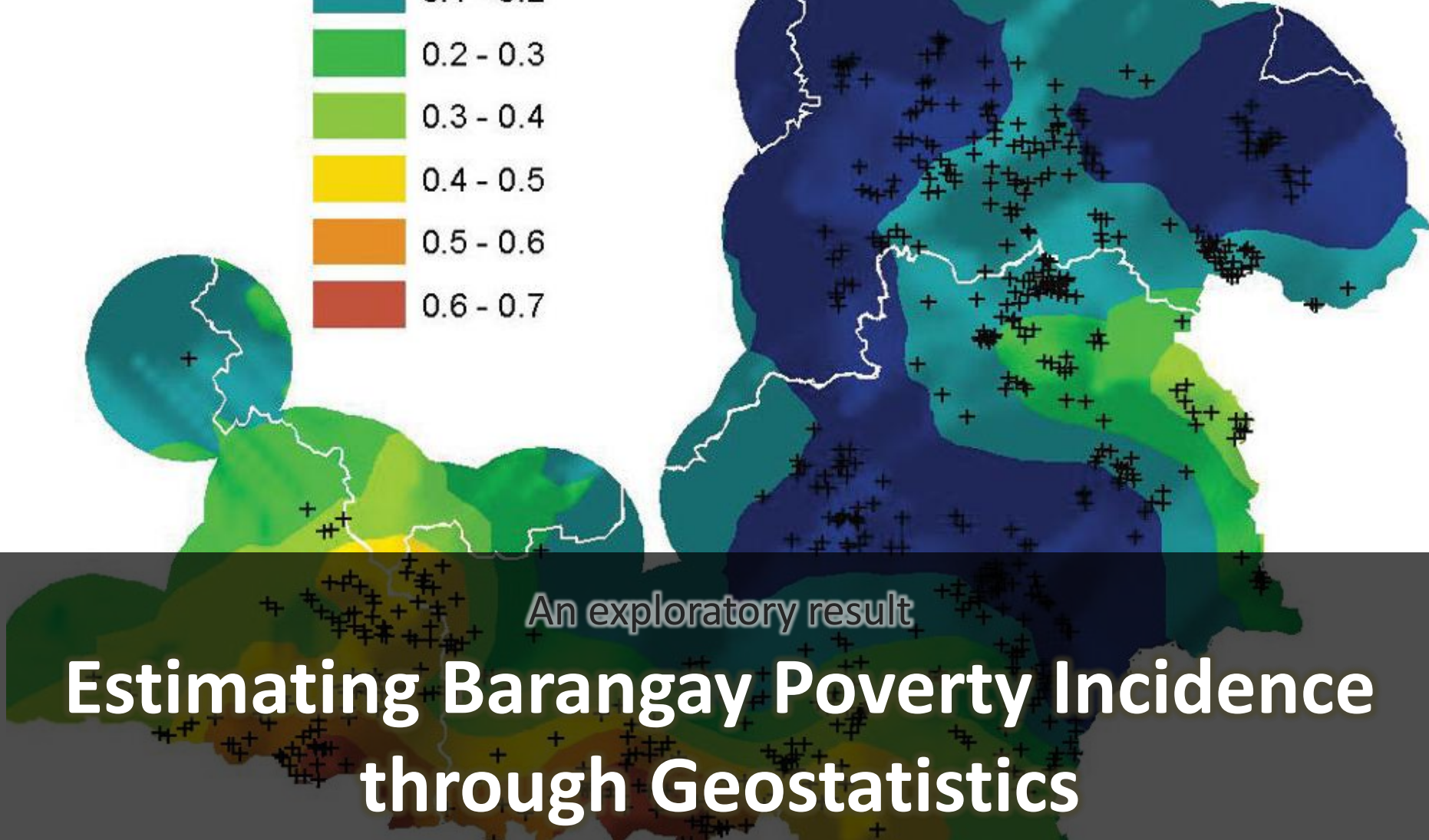




**National Anti-Poverty Commission**

Policy Monitoring and Social Technology Service (PMSTS) Unit



# Small Area Estimates of Poverty Indicators

- What we have so far are 2009 Intercensal Small Area Estimates of Poverty Indicators (poverty incidence, poverty gap, and poverty severity) produced by the NSCB from:
  - 2009 Official Poverty Incidence
  - Time invariant variables from the latest Family Income and Expenditure Survey (FIES), Labor Force Survey (LFS), and Census of Population and Housing (CPH)
- SAE methodology employed made use of the Elbers, Lanjouw, and Lanjouw (ELL) methodology

# Geographic Distance and SAE

- Let's assume that the SAE already reasonably takes into account time invariant variables.
- We can hypothesize that further differentiation can already be attributed to geographic distance.
  - As shown by PMSTS in an earlier Spatial Autoregression (SAR) Study, poverty at the municipal level is significantly spatially autocorrelated
  - Similar results had been produced by the 2012/13 Philippine Human Development Result (PHDR)

# Assuming Poverty Clustering

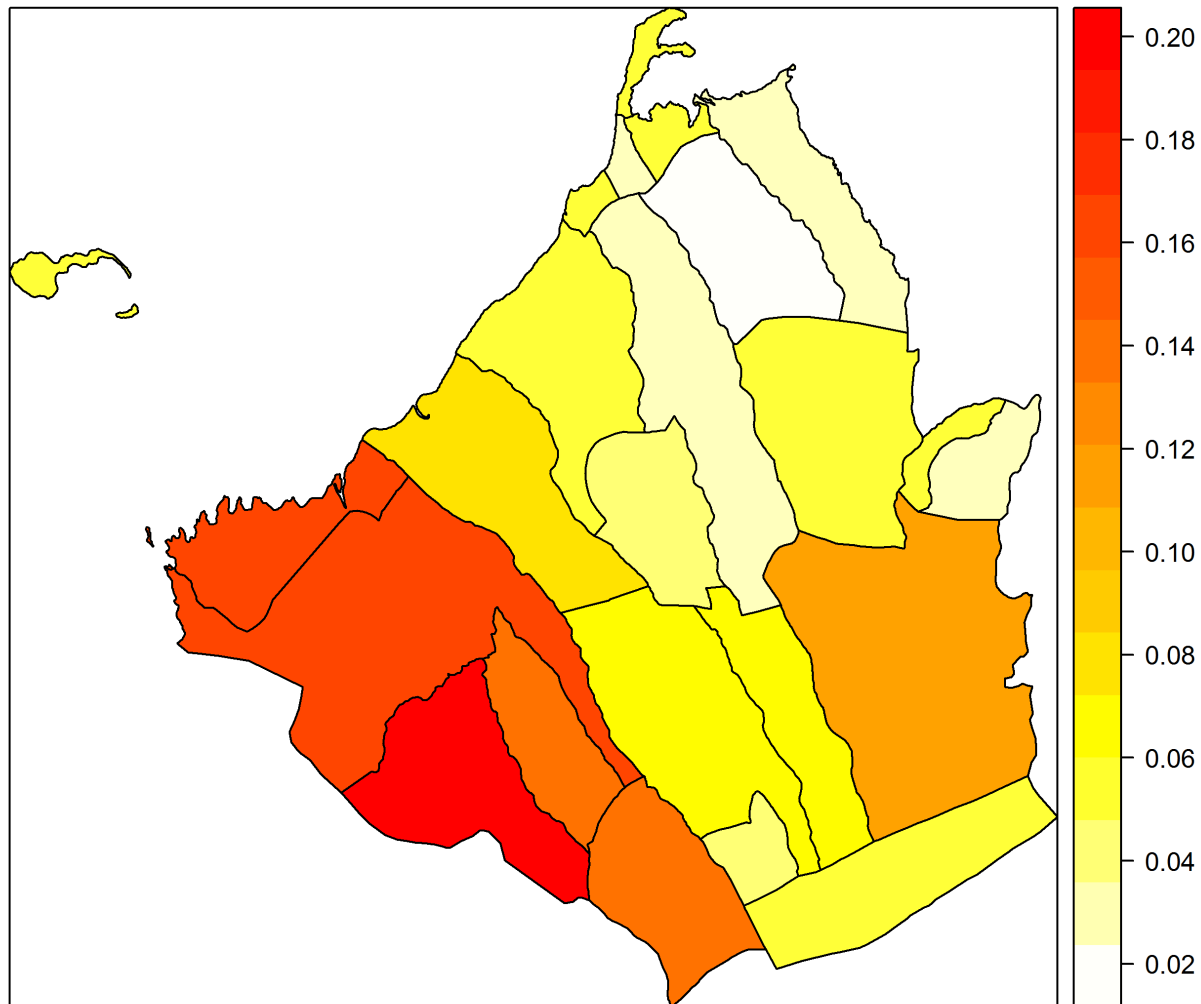
- There are various ways to test this, but the best one may be to conduct a spatial autoregression analysis on actual poverty counts.
  - But if we have this, no need for SAE to begin with.
  - We can do it through NHTS-PR, but it is not full census and may skew the results
  - We can do it through geomatics, but existing poverty morphologies in the Philippines are limited to urban areas (the Manila Observatory).
- Approach: Assume for the moment that it is true. Besides, there are various microeconomic *a priori* models that predict “poverty clustering” (income sorting urban economics models)

# Zooming in on the SAE

- Assuming that poverty at an even smaller level is determined primarily by geography, we already have several tools to estimate lower levels, say, zooming in from Municipal-level SAE to Barangay level poverty estimates.
- Tools from Geostatistics (Kriging, Semivariogram, Semivariance) – to be explained later
- The only requirement would be to do it in reasonably contiguous areas (say, at the provincial level).

# Test Case: Cavite Province

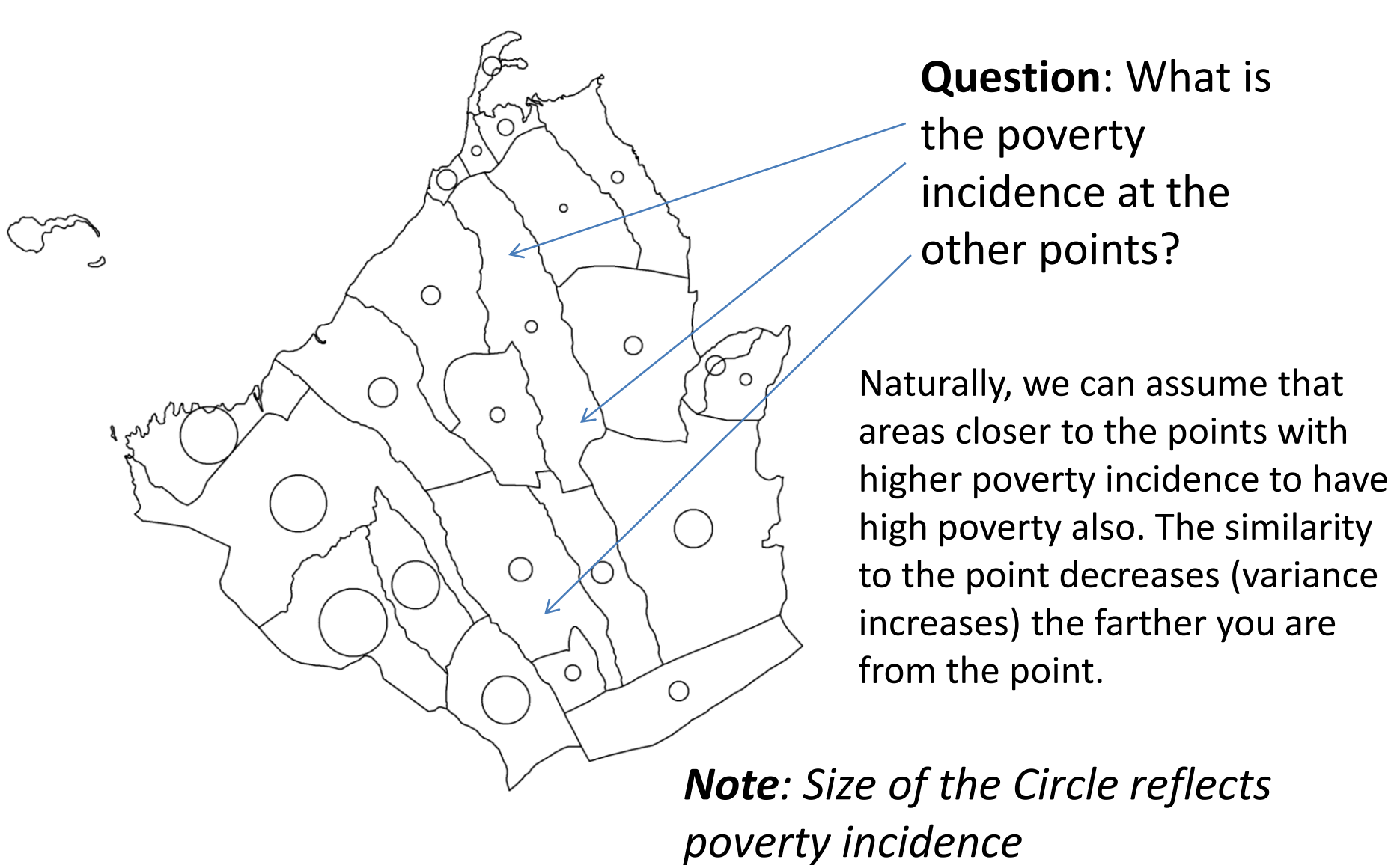
CAVITE - SAE



# From Shape to Point Data

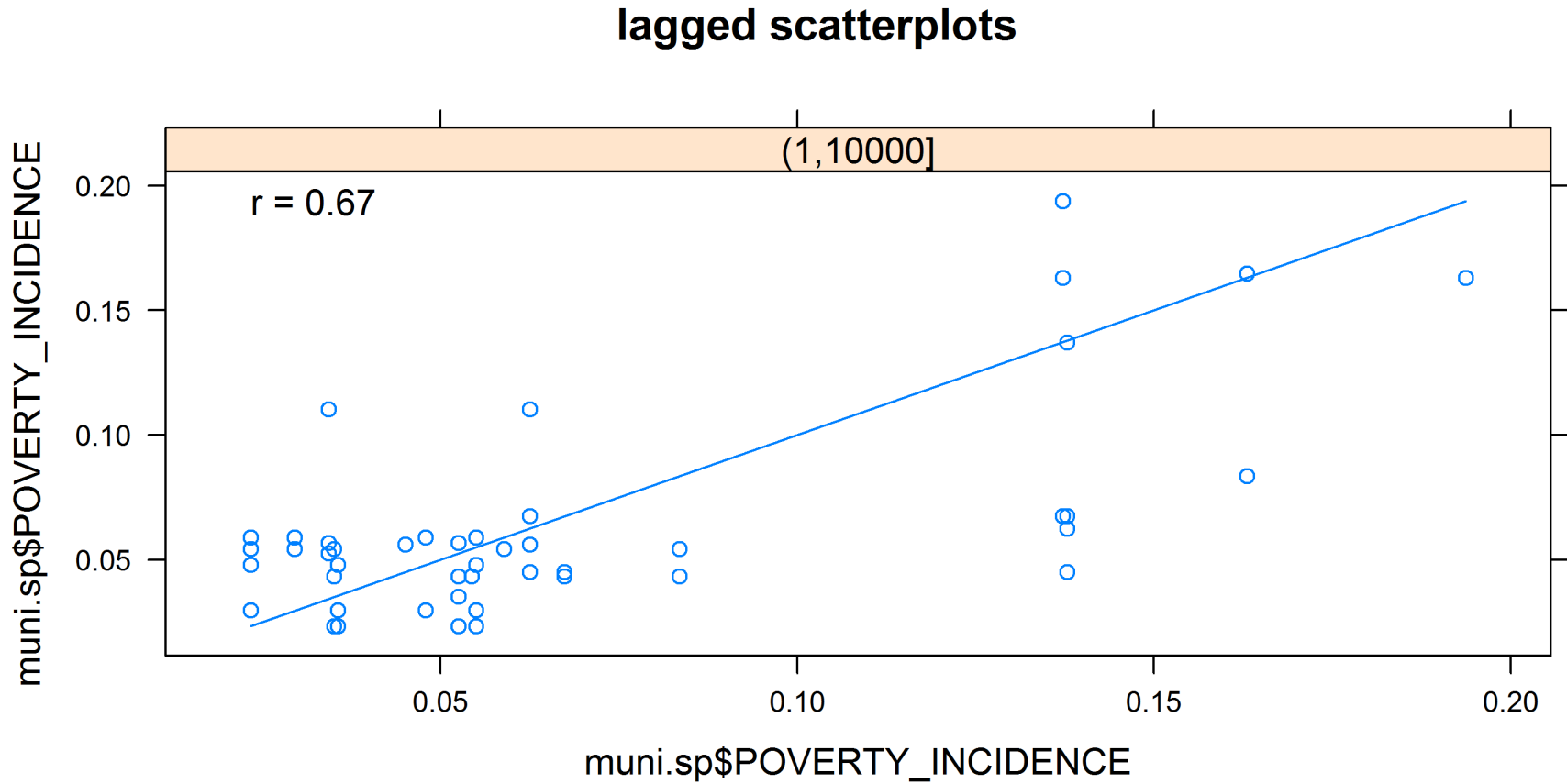
- We can suppose that these municipal level estimates are taken from the centroids of the municipalities .
  - We can adjust it later to centroids of the most populous barangay of municipalities, or other metrics.
- The problem then becomes, how does the Poverty Incidence move as we closer/away from these various points?

# From Shape to Point Data





# H-Scatterplot of Poverty Incidence



# Geostatistics

- The value of the rest of the points can be estimated through geostatistics.
  - Semivariogram Analysis and Kriging
  - First known applications are in the mining industry.
- **Semivariogram** – A graph of the variance vs. the distance, shows how the variance moves as one moves away from a point.
- **Kriging** – Fits a model curve to the semivariogram and predicts the values at the rest of the points in the map using the model curve.

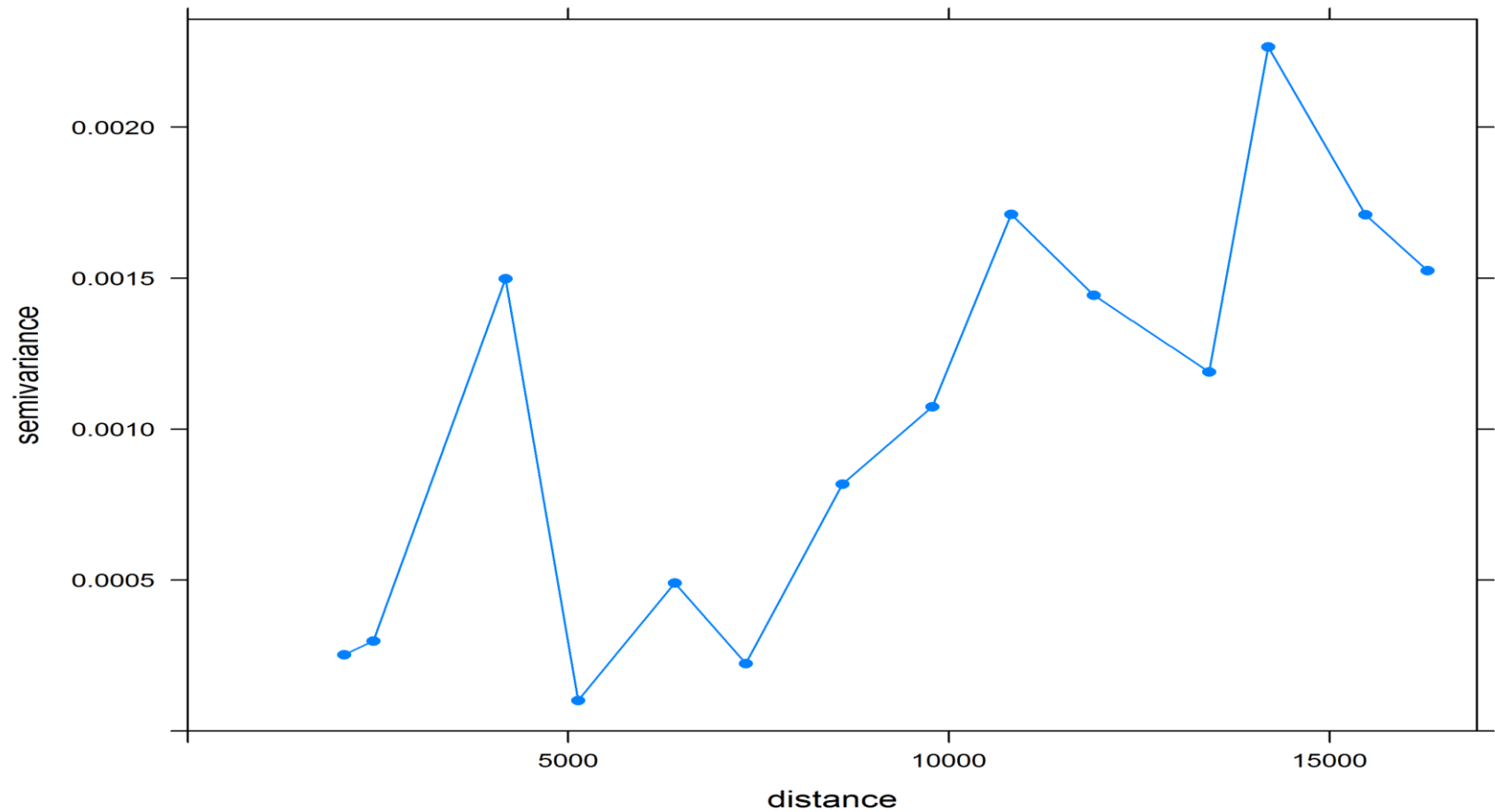
# Steps in Kriging

1. Compute the semi-variance and generate the semi-variogram.
  2. Fit the experimental variogram to the selected variogram model (nugget, spherical, exponential, Gaussian, power, etc.).
  3. Estimate Kriging weights at unknown location within some neighbourhood. To do this, one can solve the Kriging equation (simple, ordinary, kriging with a trend, etc.) to find the weights (that make the Kriging estimator UMVUE).
- *We will all be doing this through R [see attached R code and .RData].*

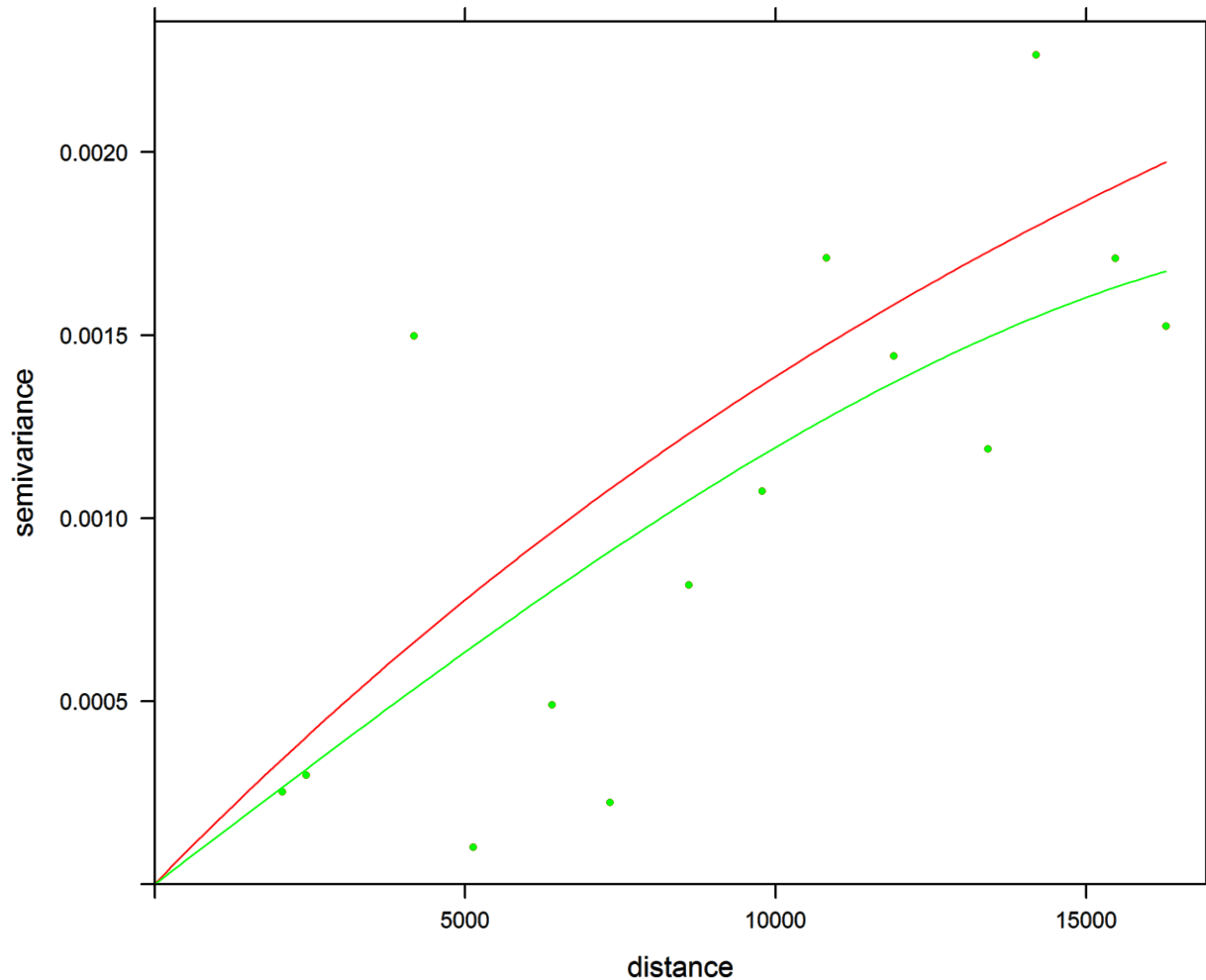
# Semivariance

- $\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i + h) - Z(x_i)]^2$ 
  - $h$  is the lag variable;
  - $N(h)$  represents the number of pairs possible, given  $h$ ;
  - $Z$  is the given function (operator);
  - $Z(x_i)$  is the variable in consideration, as a function of spatially distributed  $x_i$ ;
  - $Z(x_i + h)$  is the variable with lag  $h$ .

# Semivariogram (Poverty Incidence)



# Semivariogram – Exponential (Red) vs. Spherical (Green) Fit



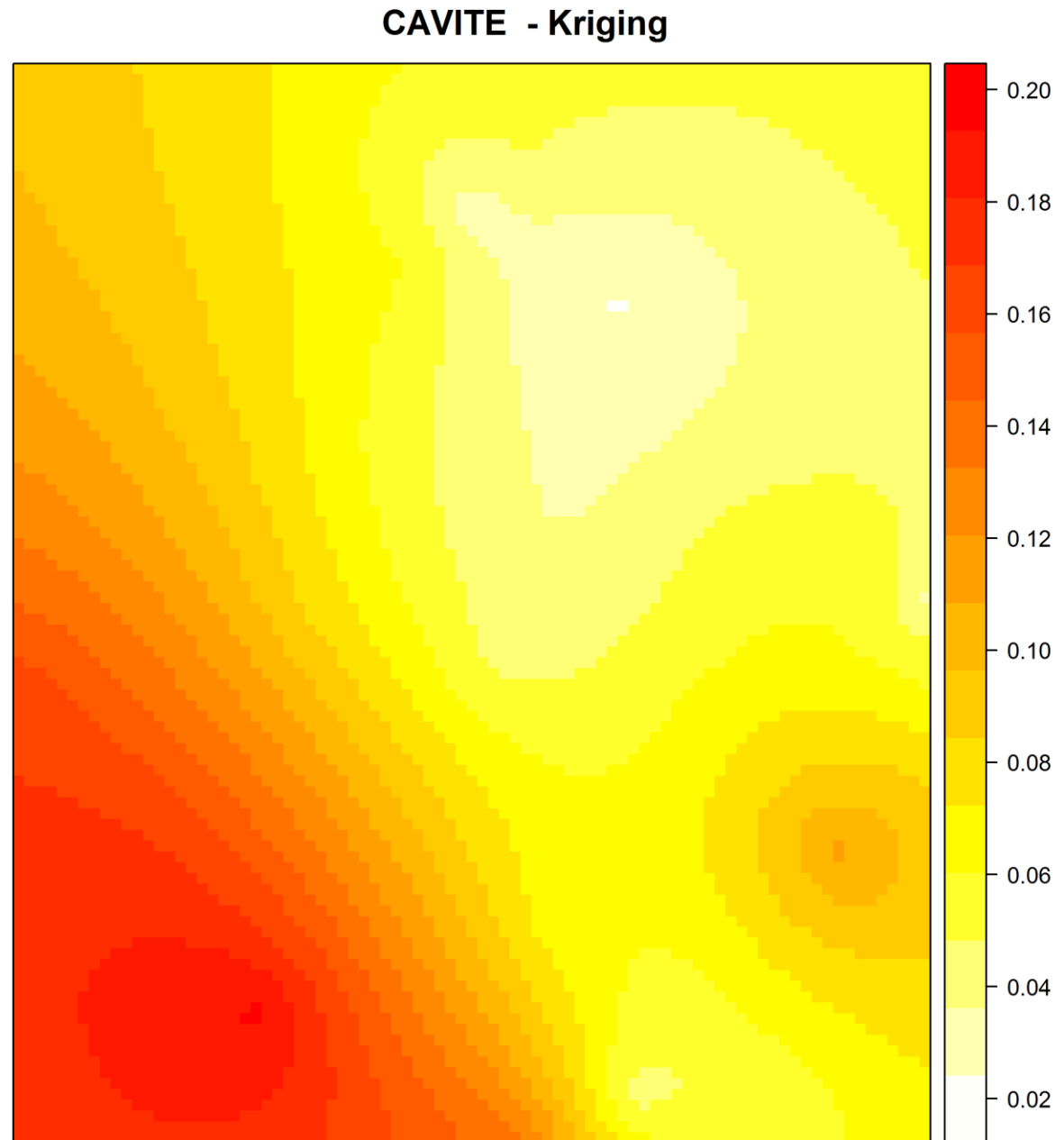
# Kriging Estimator

$$\widehat{Z(x)} - \mu(x) = \sum_{i=1}^{n(x)} \lambda_i [Z(x_i) - \mu(x_i)]$$

- a basic linear regression estimator  $\widehat{Z(x)}$ , which estimates random variable  $Z(x)$
- $x$  and  $x_i$  are vectors for location at the point of estimation, with  $x_i$  as indexed neighbor of  $x$ ;
- $\mu(x)$  is the mean of  $Z(x)$ ,  $\mu(x_i)$  is the mean of  $Z(x_i)$  i.e. the mean of neighbours of  $x$ ;
- $n(x)$  the number of data points defined as part of  $x$ 's neighborhood;
- $\lambda_i$  is the Kriging weight corresponding to  $z(x_i)$ ; These are derived from covariance function describing the residual  $Z(x) - \mu(x_i)$ .

# Heat Map from the Kriging Estimator

- Note that the model accurately predicted two “peaks” of poverty.



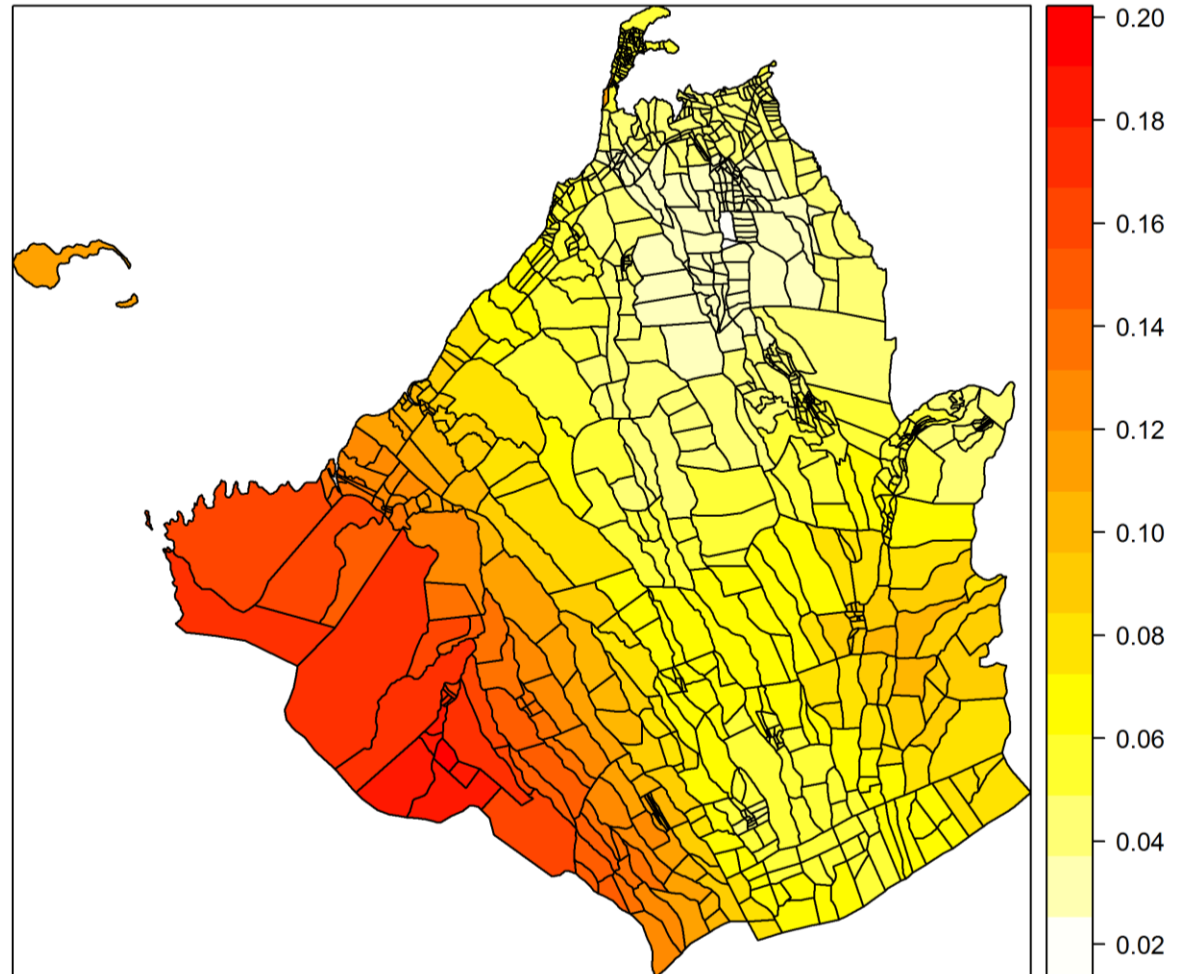


# Polygon/Zonal Statistics

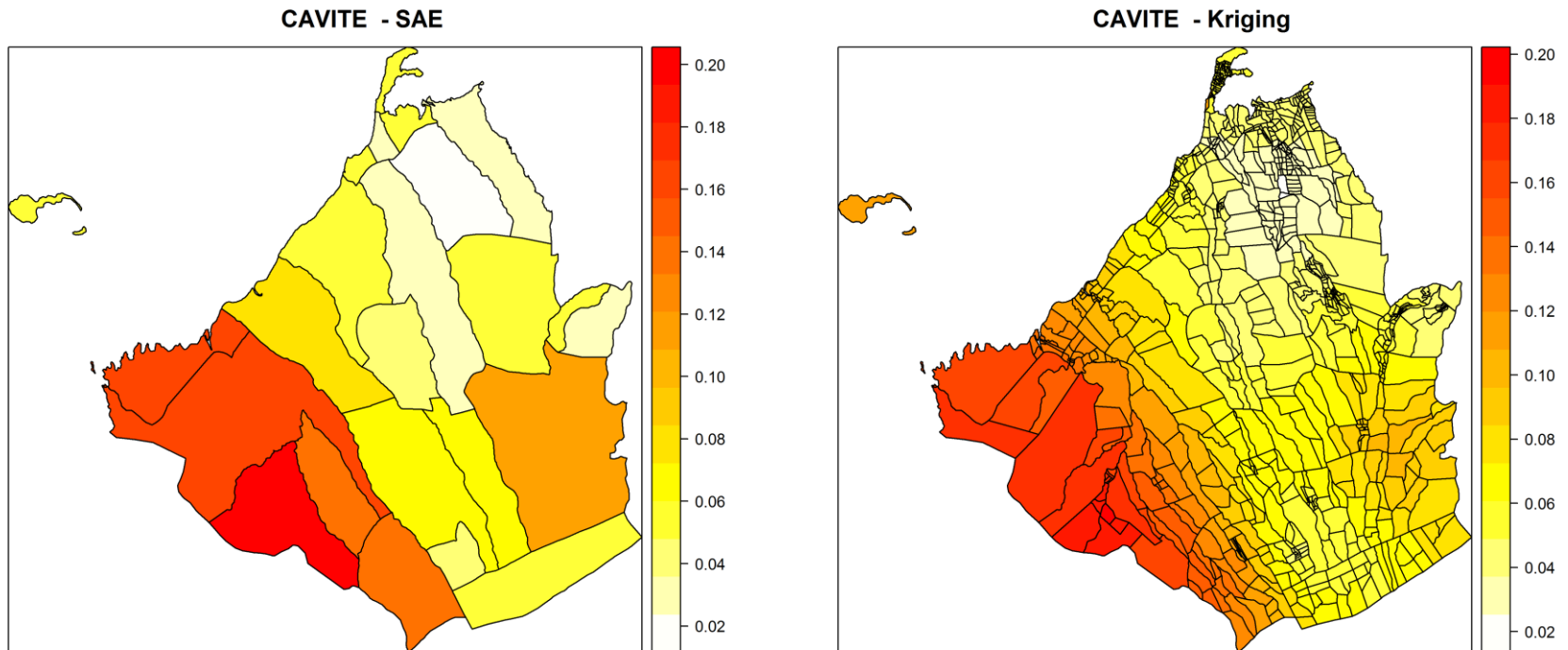
## Map for Cavite

CAVITE - Kriging

- It is no longer necessary that Barangays within a municipality has the same poverty incidence.
- See attached document (in .xlsx) for the actual predicted poverty incidence.



# Comparing the Kriged Estimates (Barangay) and the SAE (Municipality)



Note that there are now gradations within municipalities. Colors of barangays depend on the poverty incidence not just of its own municipalities but also neighboring ones.

# Next Steps

- Use a different centroid (most populous barangay, or barangay with highest household poverty incidence in a municipality [from NHTS-PR, if available])
- Co-Kriging with other municipal-level variables which are known determinants of poverty – population (mostly from NHTS also)